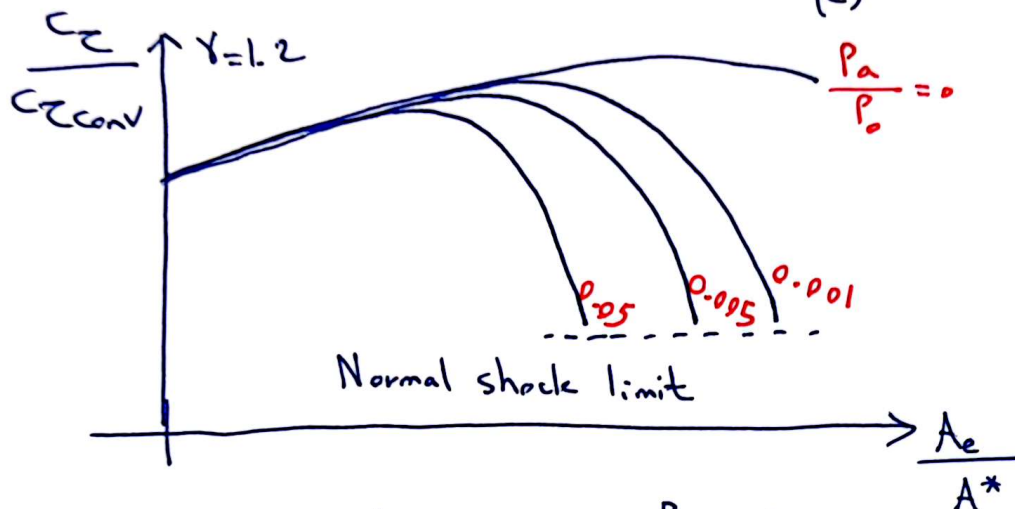


Friday - 28-2-2025 : Thrust coefficient vs. Area-ratio (Sec 6)

Q Define  $\frac{\tau}{\tau_{conv}} = \frac{C_\tau}{C_{\tau_{conv}}}$  & plot  $\frac{C_\tau}{C_{\tau_{conv}}}$  vs  $\left(\frac{A_e}{A^*}\right)$

$C_\tau : (xxx)$   $C_{\tau_{conv}} : (C) \rightarrow \frac{(xxx)}{(C)} \text{ vs } \frac{A_e}{A^*}$



Curves correspond to various  $\frac{P_a}{P_o}$  (ambient to combustion chamber pressure)

How to plot a line in this graph?

1) Take an  $\frac{P_a}{P_o} = 0.05$  &  $\gamma = 1.2$

2) Calculate  $C_{\tau_{conv}}$  for the given condition in (1) from Eq.

(C)

(3) For  $\frac{A_e}{A^*} = 1$  to  $100$

Find  $M_e$  from  $\frac{A_e}{A^*} = f(M_e)$  relation (sup. Mach)

Use  $M_e$  to calculate  $\frac{P_e}{P_o}$  from isentropic relation

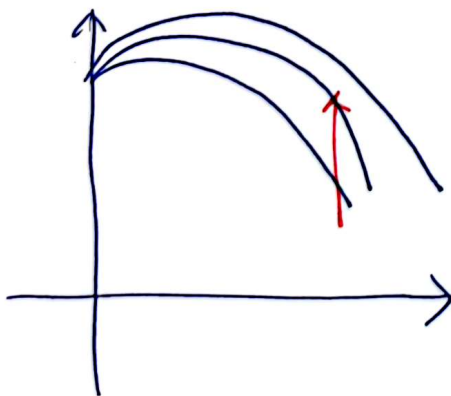
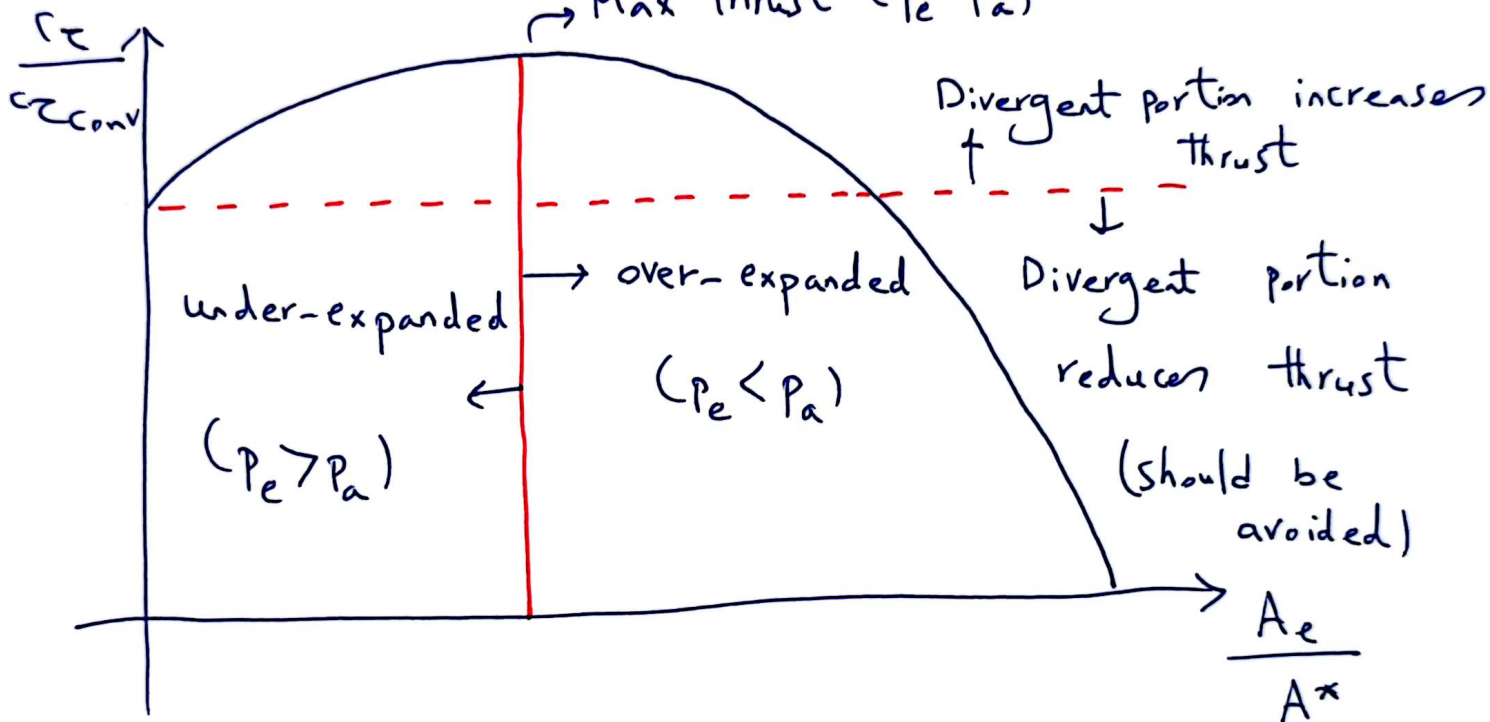
(P-11)

Calculate  $c_c$  from (xxx), compute  $\frac{c_c}{c_{c_{conv}}} = \frac{(\text{xxx})}{(c)}$

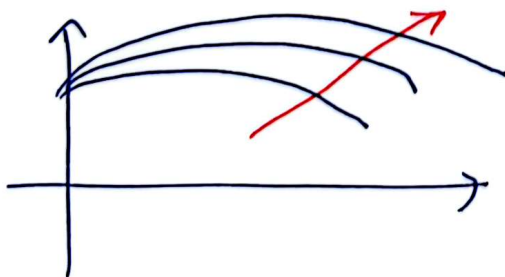
plot  $\frac{c_c}{c_{c_{conv}}}$  vs  $(\frac{A_e}{A^*})$ , do not show negative numbers

End

plot x-axis in log-scale



"Decreasing back (ambient) pressure or increasing altitude"  
over (The vertical line)



To keep  $\frac{c_c}{c_{c_{conv}}}$  maximum with altitude,  $\frac{A_e}{A^*}$  must increase

END

chemical ...

When a rocket flies, it goes from high  $\frac{p_a}{p_o}$  to low  $\frac{p_a}{p_o}$

,  $p_o = \text{cte}$ ,  $p_a \downarrow$  as altitude  $\uparrow$

If  $\frac{c_z}{c_{z_{\text{conv}}}} < 1$ , divergent portion is hindrance.

Actual rocket never operate in  $\frac{c_z}{c_{z_{\text{conv}}}} < 1$

Example: Problem 11.2 of textbook

An ideal rocket thrust chamber has  $p_o = 1.0 \text{ MPa}$  &  $T_o = 3000 \text{ K}$ ,  
 $\gamma = 1.2$ ,  $\bar{M} = 25$ ,  $A_t = 0.1 \text{ m}^2$

Calculate  $M_e$ ,  $u_e$ ,  $p_e$ ,  $\dot{m}$ ,  $c^*$ ,  $c_z$  for following cases:

① Convergent nozzle ( $p_a = 0$ )

② C-D nozzle ( $A_e = 4.06 \text{ m}^2$ ),  $p_a = 0$

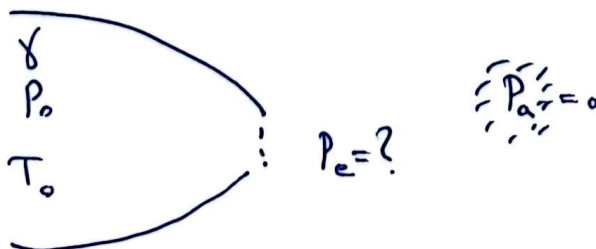
③ Convergent nozzle,  $p_a = 0.1 \text{ MPa}$

④ C-D nozzle,  $A_e = 4.06 \text{ m}^2$ ,  $p_a = 0.1 \text{ MPa}$

Solution:

$$R = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{25} = 332.6 \frac{\text{J}}{\text{kg K}} \quad \gamma = 1.2$$

Case ①



(P-3)

Strategy to solve for convergent nozzle:

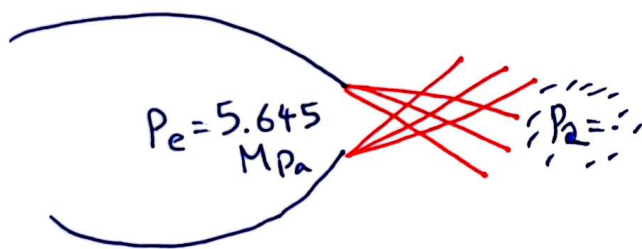
Let's assume  $P_e = P_a = 0$ ,  $P_0 = 1.0 \text{ MPa}$

$$\frac{P_0}{P_e} = \frac{1.0 \text{ MPa}}{0} = \infty \rightarrow \frac{P_0}{P_e} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}} = \infty \Rightarrow M_e = \infty$$

Wrong!

As our assumption failed, now assume  $M_e = 1$

online calc.  $\rightarrow \frac{P_e}{P_0} = 0.5644 \rightarrow P_e = \frac{P_e}{P_0} \times P_0 = 0.5644 \times 1.0 \text{ MPa}$   
 $= \underline{5.645 \text{ MPa}}$



Expansion waves occur to drop the pressure.

$$\frac{T_e}{T_0} \text{ (online Calc.)} = 0.9091 \rightarrow T_e = \frac{T_e}{T_0} \times T_0 = 0.9091 \times 3000 = \underline{2727 \text{ K}}$$

$$u_e = M_e \sqrt{\gamma R T_e} = 1043 \frac{\text{m}}{\text{s}}$$

$$\rho_e = \frac{P_e}{R T_e} = \frac{5.645 \times 10^6}{332.6 \times 2727} = \underline{6.224 \frac{\text{kg}}{\text{m}^3}}$$

For convergent nozzle,  $A_e = A_t = 0.1 \text{ m}^2$

$$\dot{m} = \rho_e u_e A_e = \underline{649.2 \frac{\text{kg}}{\text{s}}}$$

$$\tau_{\text{conv}} = \dot{m} \times u_e + (P_e - P_a) A_e = (649.2 \times 1043 + (5.645 - 0) \times 0.1 \times 10^6) = \underline{1.242 \text{ MN}}$$

$$C^* = \frac{P_0 A^* = A_e}{m} = \frac{10 \times 10^6 \times 0.1}{649.2} = \boxed{1540 \frac{m}{s}}$$

$$C^* = \sqrt{\frac{1}{\gamma} \left( \frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{\gamma-1}} R T_{0_2}} = \boxed{1540 \frac{m}{s}}$$

} Same  
answer

$$C_{\tau_{conv}} = \frac{\tau}{P_0 A^*} = \frac{1.242 \times 10^6}{10 \times 10^6 \times 0.1} = \boxed{1.242}$$

} Same  
answer

$$C_{\tau_{conv}} \text{ from } \textcircled{C} = \boxed{1.242}$$